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FEATURES

omniauth and google apps

Matt Aimonetti

Today I struggled to get OmniAuth and Google apps to work properly together. I just wanted to add authentication to my application and restrict access to only my Google Apps domain users. I was hoping it would be straight forward since I could use Google's OpenID service.

Turns out it wasn't that hard, but the lack of documentation made me lost a couple hours. I therefore updated OmniAuth's wiki and wrote this quick post so hopefully you won't waste time looking for simple details.

You actually only need to add 2 gems to your Gemfile:

```
gem 'omniauth-openid'
gem 'ruby-openid-apps-discovery'
```

Now, the second gem is the one I didn't know about. The gem is actually provided by Google itself. It turns out, Google Apps use a custom discovery protocol. They monkey patched the popular OpenID Ruby libraries so you can just drop in their gem and their discovery system will magically work.

You need to require 4 files to get everything setup properly:

```
require 'omniauth-openid'
require 'openid'
require 'openid/store/filesystem'
require 'gapps_openid'
```

The first one is the omniauth extension for OpenID, the second one is the main Ruby OpenID library (needed so we can set our SSL cert). The third one allows us to store temporary data on disk instead of keeping it in memory (optional). And finally, the last one is Google's magical gem to get their discovery system working.

Because we are going to communicate via SSL, we want to make sure that the OpenID library uses our certs to verify the SSL communications:

```
OpenID::Fetcher.class_eval do
  ca_file = "/absolute/path/to/ssl_cacert.pem"
end
```

(Obviously, you need to change the path to your own cert)

We are almost done with the setup, we just need two more things:

make sure you are using a session.

setup OmniAuth

I'm using Sinatra, so I'll load the Rack::Session middleware before I set OmniAuth:

(Rails turns that option by default, so you don't need to worry about it)

Then I can finally setup OmniAuth:

There are two important things to notice. First, because I set the provider's name to be 'admin', the magical paths provided by OmiAuth will use that name (/auth/admin). Secondly, and more importantly, notice how I added the name of my Google Apps domain at the end of the identifier:

```
'https://www.google.com/accounts/o8/site-xrds?hd=' +
your_domain_name
```

In Sinatra, you just need to define the routes OmniAuth would use (same goes for Rails, just use the router for that).

By default, omniauth now offers you a /auth/admin endpoint that will push the user through Google Apps authentication. Once the authentication is over, the user will be redirected to the following endpoint:

```
# Callback URL used when the authentication is done
post '/auth/admin/callback' do
  auth_details = request.env['omniauth.auth']
  session[:email] = auth_details[:email]
  redirect '/auth/admin/welcome'
end
```

You can access the authentication details from request.env['omniauth.auth'] and redirect the user to another page, like the admin landing page for instance.

```
get '/auth/admin/welcome' do
  if session[:email]
    erb :welcome_boss
  else
    redirect '/auth/admin'
  end
end
```

On the landing page, you need to verify that the user is logged in, in this case, during the previous step, we added the email of the user to her session. We can therefore verify the presence of that information to confirm the authentication status. If the user is authenticated, then we'll render an ERB template otherwise we'll redirect her back to the login page.

We should also provide an endpoint in case the authentication failed:

```
get '/auth/failure' do
  params[:message] # do whatever you want here
end
```

Note that by default, in dev mode, OmniAuth won't redirect the user there. To enable this behavior, use the following snippet (works with any rack app, Rails, Sinatra or whatever):

```
OmniAuth.config.on_failure = Proc.new { |env| OmniAuth::FailureEndpoint.new(env).redirect_to_failure }
```

Using Google Apps for authentication with OmniAuth is trivial as long you know two things:

the identifier url: 'https://www.google.com/accounts/o8/site-xrds?hd=' + your_domain_name

Google's discovery service gem

This blog post was written using omniauth 1.1.1, omniauth-openid 1.0.1, rack-openid 1.3.1 and ruby-openid-apps-discovery 1.2.0. This might not apply to you if you come from

meet the merbists lori holden

Matt Aimonetti

](<https://loriholden.com/>)

Today, I'm interviewing Lori Holden from YellowPages.com / AT&T Interactive .

I met Lori on IRC a bit before MerbCamp 2008 where she gave a very interesting talk on Sequel .

Matt Aimonetti Hi Lori, could you please introduce yourself and the company you work for?

Lori Holden: Hi Matt, my name is Lori Holden. I originally became interested in software development when I was 10. Since then, I have been a professional developer for over 10 years now and currently work as a Senior Software Engineer for AT&T Interactive.

Matt Aimonetti How did you get started with Ruby, what's your background?

Lori Holden: Years ago I had been hearing talk about an 'upcoming' new language called Ruby. At the time it didn't have many people in the US working with it, but I really loved how similar to Smalltalk it was.

My core background experience is centered around Ruby, Java, C++, PHP, and Javascript. Outside of that, I consider myself a language geek... and have quite a bit of experience in Python, Smalltalk, OCaml, IO Language, and Perl, along with several others.

Matt Aimonetti You chose to learn, support and use Merb, could you please let us know why and how that happened?

Lori Holden: I was one of the original Rails developers. Since then, I have used Rails off and on... but have found myself growing more and more unsatisfied with it. On top of that, I have also found myself growing out of touch with the core developers over the last couple years, which I am sure doesn't help things. Merb is fresh. I find it carries a lot less baggage with it than Rails. Even better, the project is still small and I seem to get along quite well with the other developers on the project.

Matt Aimonetti You have been contributing to the Merb project and gave a Sequel presentation during MerbCamp 08, what is your motivation?

Lori Holden: Primarily, I just like the developers. Everyone seems to be really passionate about Merb and are willing to push it in some really cool directions. At the same time, Merb still manages to remain slim and with very little bloat. I really enjoy helping projects push through to release and I was glad to be a part of that for Merb. Sequel is a really wonderful tool, and I was finding that just not that many people in the Merb community were using it. I had a lot of experience with getting Merb and Sequel to work with

each other, and figured I should share my experience with everyone else.

Matt Aimonetti: What is your favorite aspect of the Merb framework?

Lori Holden: This is a really odd one, but my favorite aspect of Merb is that I know it very well inside and out. If I need to browse the source for some feature, I already know where to go and what to look at. I attribute this to how slim the merb-core has remained. I think it is quite easy to work with.

Matt Aimonetti: Could you please mention an aspect of Merb you hope to see be improved in the near future?

Lori Holden: Merb is still immature in some areas. Gem management is one thing that I am finding to be a bit lacking. The support for Sequel could use a bit more improvement... but I suppose that's something I should work on. And finally, something that I hope Merb always works on: Keep it Small. Keep it Simple. Keep it Core. Bloat should be a plugin away, not an upgrade.

Matt Aimonetti: Thank you for your time. Anything else you would like to add?

Lori Holden: Thank you Matt. I hope you and your wife have a great winter solstice.

retrospective on freelancing consulting

Matt Aimonetti

Being a consultant is great, being a consultant working exclusively with Ruby is awesome, being a Ruby consultant using Agile methods on great projects with cool clients is just super-awesome. The English language doesn't seem to have a word/expression defining this feeling without using the "f-word" (at least, there's nothing I can think of right now) I have to say that I owe a lot to the Ruby community and especially the Rails core team. Without you/them, I wouldn't be able to do what I do and I would certainly not have so much fun.

Enough blahblah, a retrospective wouldn't be a retrospective if I wouldn't look at ways to improve the process.

Looking at back on 2007, I'll look at what I did good and where I could have done better.

communication wise , I think things worked out quite well.

Daily stand up

The daily stand up/scrum proved to be very efficient on all projects. it usually takes a short adaptation time on the clients side, but once everybody is on the same page, meetings are short and efficient .

Clients adapted more or less well to the whole Iteration process . On some occasions, I ended up being in charge of managing iterations, confirming my suggestions with the client during our iteration planning. I wish I was able to get all my clients to fully understand and enjoy the process, but oh well, the reality is that some clients just want the work done and don't care too much how you do it. In general iteration planning sessions went quite well, they really helped us defining priorities and manage the budget. However, I feel the need to bring more structure into these meetings .

Retrospectives/demos could really be improved. I have to admit that I was not able to organize a single great retrospective with my clients. That's one major draw back of working remotely and being a contractor. Since I would deploy few times a week, the weekly demo didn't always made sense. I therefore decided to use Jing and create a weekly screencast of the iteration changes. This way my clients know exactly what was done and can test whenever they want. We still have our demos from time to time especially when major new features are implemented. In 2008, I really want to improve clients' retrospective .

While lighthouse helped me keeping track of my iteration user stories, unfortunately, I noticed that clients got confused when using the same tool to report bugs and to manage iterations. I'll still be using lighthouse for bug reports but or I will have to finish my agile project manager or I'll use Josh Knowles' of he finally releases his software ;)

Bdd/Rspec was a big winner in '07. People might argue about the real benefit of RSpec vs test unit. Well, I can

only talk about my own experience, after using RSpec for a year, the way I write my applications totally changed. TDD certainly became a reality. Even my relationship with my clients changed. I now get my clients to write my tests, well, not exactly but almost. I get them to define the expected behaviors using the usual it..should..do syntax. I then just need to transpose my clients expectations in contextualized specs. once the iteration is over, I output a spec report that i give to my client. I never had so much fun writing tests!

One thing I need to focus on in 08: ajax & js tests!

Talking about JavaScript, I finally found a way to write Js and have fun. Lowpro totally changed my view of Js and I can't wait to push things further.(my December talk on Unobtrusive Javascript should be online sometimes soon)

Talking about pushing things further, I need to see if I can totally dish svn and only use git + gitosis. if you didn't check Git yet, go check on the awesome peepcode put together by topfunky .

Misc things that didn't work that well:

unreliable subcontractors

hard to find designers

having to say no to so many cool projects (always a good problem to have though)

Various things that I enjoyed:

Joyent accelerators

twitter

sd.rb

autotest

iPhone (and probably a lot other stuff that I can't remember right now)

Major changes planned for 08:

pairing with another local developer complementing my skills.

better weekly retrospectives

tech lunch at least once a month

learn Erlang or another programming language

Top 10 AI Tools in 2023 That Will Make Your Life Easier

Unknown · TechCrunch

In this article, we explore the top 10 AI tools that are driving innovation and efficiency in various industries. These tools are designed to automate repetitive tasks, improve workflow, and increase productivity. The tools included in our list are some of the most advanced and widely used in the market, and are suitable for a variety of applications. Some of the tools focus on natural language processing, such as ChatGPT and Grammarly, while others focus on image and video generation, such as DALL-E and Lumen5. Other tools such as OpenAI Codex, Tabnine, Canva, Jasper AI, and Surfer SEO are designed to help with specific tasks such as code understanding content writing and website optimization. This list is a great starting point for anyone looking to explore the possibilities of AI and how it can be applied to their business or project. So let's dive into

1. ChatGPT

ChatGPT is a large language model that generates human-like responses to a variety of prompts. It can be used for tasks such as language translation, question answering, and text completion. It can handle a wide range of topics and styles of writing, and generates coherent and fluent text, but should be used with care as it may generate text that is biased, offensive, or factually incorrect.

Pros:

Generates human-like responses to a variety of prompts

Can be fine-tuned for specific tasks such as language translation, question answering, and text completion

Can handle a wide range of topics and styles of writing

Can generate coherent and fluent text, even when completing a given text prompt.

Cons:

May generate text that is biased or offensive

Can generate text that is not accurate or factually correct

May require large amounts of computational resources to run

The model can sometimes generate text that is not coherent or fluent, depending on the prompt given.

Overall, ChatGPT is a powerful tool for natural language processing, but it should be used with care and with an understanding of its limitations.

2. DALL-E

DALL-E is a generative model developed by OpenAI that is capable of generating images from text prompts. It is based on the GPT-3 architecture, which is a transformer-based neural network language model that has been trained on a massive dataset of text. DALL-E can generate images that are similar to a training dataset and it can generate high-resolution images that are suitable for commercial use.

Pros:

Generates high-resolution images

Can generate images from text prompts

It can be fine-tuned for specific tasks such as generating images of a certain style or category

Cons:

May generate images that are not entirely original and could be influenced by the training data

May require significant computational resources to run

The quality of the generated images may vary depending on the specific prompt

Overall, DALL-E is a powerful AI-based tool for generating images, it can be used for a variety of applications such as creating images for commercial use, gaming, and other creative projects. It is important to note that the generated images should be reviewed and used with care, as they may not be entirely original and could be influenced by the training data.

3. Lumen5

Lumen5 is a content creation platform that uses AI to help users create videos, social media posts, and other types of content. It has several features that make it useful for content creation and marketing, including:

Pros:

Automatically summarizes text from a blog post, article, or other source into a script for a video or social media post

merb progress

Matt Aimonetti

As they say: fail early, fail often. Well, I've been failing to blog recently, but as always I have some good excuses ;)

Yehuda has been blogging a lot about the work done on the merge.

I have been busy working on probably the awesomest CouchDB Ruby DSL/ORM .

I have been working with the Rails Activists on the new wiki and many other projects.

My sister is visiting from France :) (most of my free time is not spent in front of a computer anymore :p)

I've been playing with MacRuby (see the end of the post)

Ohh and spent some time maintaining Merb and preparing 1.0.9

Merb 1.0.9

Merb 1.0.8.1 has some issues with the merb-cache settings set in init.rb being called twice when you use some Rake tasks. (rake db:automigrate for instance). The problem is due to the fact that the rake tasks load the dependencies twice:

Merb rake file loads all of your app dependencies and their rake tasks. (for instance rake db:automigrate is a rake task coming from merb_datamapper)

The task you are invoking might start merb itself to load the models etc.. and starting Merb reloads the dependencies.

Dependencies have the option to have a require block. A require block is a Proc that gets called when the dependency is being required. In the case of 1.8.1, we added a default block to set merb-cache. The problem was that the block was being set and called twice and merb-cache was complaining that the default cache was already setup.

To fix this issue we worked on a band-aid type solution (read: kinda evil but ok). Even if you start Merb multiple times, the init.rb file will no only load once and the dependency require blocks will only be called once. However, in the case of merb-cache, the setup block is being called everytime you go through the bootloader. That's why we added a verification on the block itself. In the long term, we are going to fix things nicely and optimize the way merb-cache loads.

We also addressed some other issues. Some people have been pointed out issues with some merb-helpers and patches were provided by the community (thanks a lot).

Talking about patches, we also accepted a patch fixing a small problem with merb-auth and some open-id servers.

Ruby 1.9.1: We started going through all the different places where things are not compatible yet, most of the work that needs to be done right away is focused on the router. Carl has been working on that and you can check his branch to see how we are doing. During the next week we will try to get things ironed out, we still don't know how we are going to deal with the fact that action-args uses parsetree which is not 1.9.1 compatible.

The upcoming version also gets a brand new feature: memory monitoring by the merb master process. The master process checks that the workers don't use too much memory (you can set what you consider beign too much memory using Merb:: Config[:max_memory]) and if one of the workers reaches the limit, it does a kill -1. It then waits a configured amount of seconds (defaults to 5) then kill using -9.

Finally, we also fixed a bunch of tiny issues.

1.0.9 should be released in the next few days.

MacRuby

Finally, because for those interested in MacRuby here are my slides from last night's SDRuby meetup:

MacRuby - When objective-c and Ruby meet

View more presentations from Matt Aimonetti .

Even if you start Merb multiple times, the init.

Matt Aimonetti

sharing rails sessions with non ruby apps

Matt Aimonetti

I wanted to share sessions between my Rails and Go applications. I wanted to let an authenticated Rails user make JavaScript API calls to an endpoint written in Go. How hard could it be?

Since I own both apps, I thought it would be as simple as sharing the secret session key and re-implementing Rails crypto process in Go. It turned out to be a lot more interesting.

In a nutshell, here is what I discovered:

It's totally doable! Here is my Go package .

If you are using a version of Rails older than 4.0, you'd better upgrade ASAP!

Rails has been criticized for security issues, but the current solution has been vetted by many experts.

Rails serializes session data using Ruby Marshal which means that someone with the secret key can inject arbitrary code in the session and it will execute server side. Switch to JSON, MessagePack or other safe serialization formats.

Security is (still) hard.

Rails Cookies are Dangerous

Because Rails serializes and deserializes the session and any encrypted/signed cookies using Ruby's Marshal library, someone with the app secret can wreak havoc. They can embed arbitrary Ruby code into the cookie, submit it with a request, and the server-side deserialization will execute that code without you noticing. Granted, this requires the attacker to have the app secret, but since 99% of the apps out there have the shared secret in their source code, anyone with access to the source code has this data. It's not data you can easily rotate when employees leave or when you are done working with contractors. Anybody with the shared secret is a potential attacker. Start by moving this data out of the code base and into an environment variable.

Rails doesn't let you change the default serializer directly. But Rails relies on ActiveSupport for its crypto work and AS supports swapping the serializer. Some people in the community are aware of this issue and monkey patch Rails to serialize their sessions using JSON or another alternative. Here is an Airbnb article and Rails 3 patch . Here is my Rails 4 monkey patch to switch the serialization to JSON. I'm using it in production with Rails 4, but it's untested on Rails 3.

You can modify either solution to use MessagePack instead of JSON if you want to fit more data in the 4K cookie size.

Once I addressed the serialization issue, I had to reimplement the crypto work done by Rails to encode and/or sign the data.

Most of us just rely on our frameworks/libraries to do the right thing, but we rarely look under the hood. I ported the logic to Golang which has an amazing support for crypto (albeit lower level than Ruby). My Go package contains an explanation of the code logic and the examples needed to decode/verify as well as encode/sign sessions that are compatible with Rails.

Here is a high level summary of what Rails does when it encodes and signs your session data:

```
key_generator = ActiveSupport :: CachingKeyGenerator .
new( ActiveSupport :: KeyGenerator . new(app_secret_key,
iterations : 1000 )) derived_secret = key_generator .
generate_key( "encrypted cookie" ) sign_secret =
key_generator . generate_key( "signed encrypted cookie" )
```

```
encryptor = ActiveSupport :: MessageEncryptor .
new(secret, sign_secret) session_content = encryptor .
encrypt_and_sign({ hello : "world" })
```

The session_content string is then set as the session cookie value. Note that you could do that in any Ruby app using ActiveSupport , making it easy to share sessions between Ruby applications (like Rails & Sinatra).

Technically, there are a lot of things going on. To avoid using the same secret to sign and encode data, Rails relies on derived keys using PBKDF2 (password based key derivation function). It treats the app secret as a password and applies a pseudorandom function 1000 times (Rails default) using a default salt. The result is a derived key so the original password isn't shared. The derived key can be regenerated identically if the salt and secret are known (because the function is pseudorandom).

The two derived keys are then passed to the MessageEncryptor class which uses MessageVerifier to do the signing. The generated keys are 64 bytes long. One key goes to the encryptor while the other goes to the verifier.

The verification is done via HMAC (SHA1) and it uses the full 64 byte key. The encoding is done via AES 256 CBC only using the first 32 bytes of the encryption derived key. Rails will only generate a 32-byte key since that's the expected key length.

The session data is serialized (using Marshal by default) then encoded via AES. Both the encoded string and the IV are encoded using base64 and joined in a string using a predefined format.

At this point, the session is encoded but it could be tampered with. To avoid that, Rails signs the encoded data using the verifier (HMAC) and appends the base 64 encoded signature to the encoded data.

presenting the rails activists

Matt Aimonetti

Today is Monday. I usually don't like Mondays. Being Monday goes with waking up early, going back to work, and lots of deadlines.

However, today is a special Monday. It's the first Monday of the year and I have a special announcement !

During the Rails/Merb merge announcement, it was mentioned that I will be joining the soon to be created "Evangelism team" .

A few people asked me what being a "Rails Evangelist" means. To reassure my parents and close friends, no, I didn't join a new cult worshipping locomotives . However, I still think that public transportation should be improved, especially in this time of crisis (but that's a different topic).

A technical evangelist, is usually someone who knows and uses a specific technology and thinks others should look into it. This is something I've been doing for Merb while being part of the core team. I initiated and helped organizing MerbCamp, re-did the wiki, started working on the merb-book, spent time looking for and listening to users, spent time with third party developers and people pushing Merb to a new level (YellowPages, Wikimedia and many others).

This interaction with the end users and the third party developers is something the entire Merb team valued a great deal and I always felt it was something the community really appreciated.

As part of the merge, it was agreed that we would push things further and have a team within the Rails team to take care of "communication". Rails is a bigger project than Merb and communication between the dev team and the users isn't always something easy to do.

That's why we have formed a separate team that will help communicate and support the community better. We now even have an official page on the Rails website itself :)

The A-Team just got announced on the Rails blog .

Instead of being called "evangelists", we are going to be called "activists". I think part of the argument was that the E-Team doesn't sound as good as the A-Team.

We started with team of 4. You might not know them yet but they all are brilliant people and I'm really glad to be working with them.

Gregg Pollack , from Rails Envy. You might remember Gregg from the Rails vs * commercials or from the Rails Envy podcasts. I've known Gregg for a little while and he's someone you can rely on and always full of energy/new ideas.

** Ryan Bates**, mainly known for his Railscasts. I only met Ryan once in person, but I've always been impressed by

his work (don't tell anyone, but I secretly dreamt of having something like Railscasts but for Merb :))

Mike Gunderloy . I actually did not know Mike but I have read and enjoyed his blog and have seen his work on the Rails guides. Mike is an experienced writer and developer. He joked the other day saying that he started programming before any member of the Rails team was even born. Mike is a great addition to the team and I'm looking forward to learning from his experience.

Gregg and Ryan also covered the event, you might want to check their blog posts (Gregg's and Mike's)

So what are we going to do?

Pretty simple. We've boiled it down to 2 sentences:

The mission of the Rails Activists is to empower and support the worldwide network of Ruby on Rails users. We do this by publicizing Rails, making adoption easier, and enhancing developer support.

if you prefer a few more details, here are some of the tasks we are going to work on:

Public Relations with media of all sizes

Ombudsman work to ensure good user-to-user support

Community Leadership at events and conferences

Media Organization to help create good promotional opportunities

Website maintenance

Documentation efforts

Developer support

Do we need help?

Absolutely! The idea is not that we are going to do all the work. The concept of this new team is to help organize the community. We are going to build a Rails Network, a network of people involved in local Rails "evangelism"/activism, people contributing and/or translating documentation, third part developers etc...

First thing would be to join the mailing list and share your suggestions, comments, concerns, etc., with us.

Secondly, we have already set up some forums to hear your feedback.

To start off, we are asking people to let us know what they would like to see happening in the Rails3 timeframe . We have other forums for more general feedback , but we need to work with deadlines so we can prioritize accordingly. Using the Rails3 milestone should help us focus on a short/medium term deadline. Long term and not specific sugges-

Real Python: How to Use Git: A Beginner's Guide

Planet Python

This tutorial shows you how to use Git to track changes in a project using just a few core commands and save clean snapshots of your work. If you've ever changed a file, broken something, and wished you could undo it, version control makes that possible. Git keeps a running history of your files so you can see what changed and when.

In this guide, you'll set up Git locally and use the core workflow from the terminal to track and record changes in a Python project. By the end, you'll have a working Git repository with a recorded commit history you can inspect and manage:

Commit History Displayed With `git log`

In the next sections, you'll create your own repository and begin building that history from scratch. Before you begin, you can download a Git cheat sheet to keep the core commands handy:

Take the Quiz: Test your knowledge with our interactive "How to Use Git: A Beginner's Guide" quiz. You'll receive a score upon completion to help you track your learning progress:

How to Use Git: A Beginner's Guide

Test your knowledge of Git basics: initializing repos, staging files, committing snapshots, and managing your project history.

How to Use Git: Prerequisites

Before you start tracking your code with Git, make sure you have the right tools in place. This tutorial assumes that you're comfortable working with the command line and have some basic Python knowledge.

Here's what you'll need to get started:

A terminal or command prompt

Python 3.10 or higher installed on your system

Note: Git and GitHub are often confused, but they're not the same thing:

Git is version control software that runs on your computer. It tracks changes to your files and manages your project's history locally.

GitHub is an online platform for hosting Git repositories. It provides collaboration tools that make sharing code, working with teams, and backing up your projects easier.

You don't need a GitHub account to use Git or follow this tutorial. Later, if you want to share your code with others or back it up online, you can optionally push your Git repository to platforms like GitHub, GitLab, or Bitbucket.

To learn more about the differences between Git and GitHub, check out [Introduction to Git and GitHub for Python Developers](#).

With these prerequisites in place, you're ready to begin setting up Git and tracking changes in your project. In the next step, you'll install Git, prepare your existing Python files, and initialize your first repository.

Step 1: Install Git and Prepare Your Project

To start, you'll check whether Git is installed on your system, prepare a simple project, and initialize a Git repository so you can begin tracking changes right away.

Check Whether Git Is Already Installed

Before you can start using Git, you need to make sure it's installed on your machine. Chances are that Git is already present on your system. To check whether Git is installed, run this command:

```
$ git --version
```

If this command displays a Git version, you're good to go and can create a project directory. Otherwise, you need to install Git on your system before continuing.

Install Git on Your System

Luckily, Git provides installers for Windows, macOS, and Linux on its official website, offering a straightforward way to install Git on your machine. Because installation steps vary across operating systems, this guide links to the official documentation rather than reproducing those steps here.

If you prefer a graphical interface, you can install a Git client such as GitHub Desktop, Sourcetree, or GitKraken. These tools install Git automatically during setup.

Once installed, open your terminal and confirm that Git is available:

```
$ git --version git version 2.24.0.windows.2
```

Your Git version may appear slightly different from this example, depending on your operating system and when you installed Git. That's perfectly fine. As long as Git is installed and the command runs successfully, you'll be able to follow along with the rest of this tutorial without any issues.

Create a Project Directory

Read the full article at <https://realpython.com/how-to-use-git/> »

merb 1.0 released

Matt Aimonetti

Japanese translation

On November 7, Yehuda Katz gave a talk at RubyConf and made 3 major announcements:

Engine Yard to offer Merb support

m|a agile to offer professional training

Merb 1.0 released!

The first announcement is pretty substantial. Engine Yard has been financially supporting Merb by letting some of their staff work on Merb, fly them to different conferences and sponsoring events like Merb Camp . Engine Yard didn't yet announce the price structure but having the option for enterprise level support for Merb is just awesome .

John Nunemaker made an interesting comment during RubyConf Pivotal Party. Something special about Merb is that it was designed to fit the needs of an audience instead of trying to create a series of tools to build a specific type of website. Having Engine Yard help to finance Merb dev and offer support is very reassuring. It's something already done by many other OSS projects such as Ubuntu, MySQL etc...

The second announcement Yehuda made was about Merb Training . I'm probably pretty biased since I am at the origin of this project. I know for a fact that a lot of people were waiting for 1.0 to get started with Merb. We are also working on getting more documentation out, and 3 books are coming up. Still, the best way to learn is to sit down with people who know Merb who can teach you the way it's intended to be used.

Training will allow you to benefit a lot from being with other people who also share the same desire to master Ruby's most powerful and flexible web framework .

What's also really exciting is that Yehuda Katz , Merb's lead developer, agreed to be a tutor for the course. I can't imagine a better way to learn. Check this page for more information about the next training session or get in contact with me if you want to organize a training session for your company.

Finally, the big news was the announcement of Merb 1.0 !

Merb 1.0 went through 5 release candidates and was finally marked as final. In the last few months, the Merb team worked hard to make things easier for people who want to get started in no time.

Let's quickly look at why Merb is awesome:

Merb is Modular . Merb is not a monolithic framework. You can pick and choose what you need. Create a 1 file app "à la Sinatra" or a rich web app "à la Rails". Merb has many com-

ponents. Only requires the ones you need and save precious resources. (Merb doesn't believe in 1 size fits all)

Merb is agnostic (kinda). Because people have different needs and different beliefs, Merb won't force you to use one ORM or another. Same thing goes for the template engine or for the JavaScript library you want to use. ActiveRecord, DataMapper, Sequel, RelaxDb, Haml, Erb, Prototype, jQuery... choose which one you want and change whenever you want.

Merb can be opinionated . Merb offers a default stack using DataMapper and jQuery, authentication, exceptions and caching setup for you. However, creating your own stack is dead easy. As a matter of fact, the guys at yellowpages.com are using Merb and were talking about creating their own stack using Sequel.

Merb let you reuse your code . Borrowed from Django , Merb has something called "slices". Slices are mini apps you can run standalone or mounted within another app. A slice is a great way to write code you can reuse. Unlike plugins which extend the framework features, slices are a way to provide encapsulated content. (David Chelimsky,RSpec's author and maintainer actually said that slices were his favorite feature in Merb 1.0)

Merb has an API . You might be wondering why having an API is awesome. Well, the truth is that the Merb Team spent time marking methods public, which are guaranteed to not break until the next major release (any change to the public API will be well documented). There is also a plugin API meaning that plugin developers won't have to worry about upgrades if they stick to the plugin API.

Merb is fast. Even though Ruby the language isn't really fast and contrary to popular opinion, Ruby for the web is one of the fastest solutions out there. (Even Rails is way faster than all the mainstream PHP frameworks) And that's what Merb is proving by being one of the fastest web frameworks available on the market. Jason Seifer will be interested to know that Merb isn't scared to scale ;) As a matter of fact, Merb is going to scale even better in the next few months as we are planning to integrate Swiftply and do some totally awesome stuff to spawn/reap workers based load. (more about that in few weeks).

Matz likes Merb . Ruby's daddy, Yukihiro Matsumoto told us he likes the flexibility of Merb and the fact that the framework doesn't create a DSL on top of Ruby. He even told us that he's going to introduce his company to Merb! We were obviously very honored and for us, it validates months of work by dozens of contributors. Here is a transcript of Matz comments about Merb and the Ruby web world:

[caption id="attachment_213" align="alignleft" width="125" caption="Yukihiro Matsumoto aka Matz"] [/caption]

"Everyone outside of the Ruby community understands that we only have 1 web application framework, named Rails, but it's not true in any sense. We have several post-Rails

‘Good, I’m Glad He’s Dead.’

John Gruber · *Daring Fireball*

The sitting president of the United States, on his blog:

Robert Mueller just died. Good, I’m glad he’s dead. He can no longer hurt innocent people! President DONALD J. TRUMP

As the elderly descend further into dementia, they lose their sense of propriety and simply speak their mind. (They also get confused and think they need to “sign” their text messages and social media posts.) Say what you want about Trump’s truthfulness generally, but here, he’s just being brutally honest. Let’s keep his “Good, I’m glad he’s dead” post bookmarked for when Trump himself finally keels over — after he chokes on a hamburger or whatever it’ll be that finally does him in — and the good people of the world rejoice and celebrate.

★

Third Stage Engineering

Brendan Gregg

The real performance of any computer hardware in production is the result of the hardware, software, and tuning; the investment and sequence of these efforts can be pictured as a three-stage rocket:

I recently presented this embarrassingly simple diagram to Intel’s executive leadership, and at the time realized the value of sharing it publicly. The Internet is awash with comparisons about Intel (and other vendors’) product performance based on hardware performance alone, but the performance of software and then tuning can make a huge difference for your particular workload. You need all three stages to reach the highest, and most competitive, performance.

It’s obvious why this is important for HW vendors to understand internally — they, like the Internet, can get overly focused on HW alone. But customers need to understand it as well. If a benchmark is comparing TensorFlow performance between HW vendors, was the Intel hardware tested using the Intel Extension for TensorFlow Software , and was it then tuned? The most accurate and realistic evaluation for HW involves selecting the best software and then tuning it, and doing this for all HW options.

I spend a lot of time on the final stage, tuning — what I call third-stage engineering . It’s composed of roughly four parts: People, training, tools, and capabilities. You need staff, you need them trained

to understand performance methodologies and SW and HW internals, they need tools to analyze the system (both observational and experimental), and finally they need capabilities to tune (tunable parameters, settings, config, code changes, etc.).

I see too many HW evaluations that are trying to understand customer performance but are considering HW alone, which is like only testing the first stage of a rocket. This doesn’t help vendors or customers. I hope that’s what my simple diagram makes obvious: We need all three stages to reach the highest altitude.

globalite update

Matt Aimonetti

Today I updated Globalite, refactoring the code and fixing some issues.

I added a bunch of aliases since I was tired of typing:

```
Globalite.current_language  
= :fr
```

I shortened it to

```
Globalite.language = :fr
```

But the old method is still available so it won't break your code ;)

The major change with this update is a better support for locales. Let's say that in our application we setup the language to :fr (French) but we didn't specify a country. Remember that Globalite lets you have specific translations/localizations per country so you can display dates, prices properly.

Previously, if my locale was set to :fr-* then Globalite would try to load the fr.yml files and not fr-FR.yml.

If you decided to use a fr.yml file for your UI localization, you might have noticed that all the Rails core messages weren't translated(now you know why). This update deals with this issue and if a translation isn't found in your locale, it will look for translations in your language but from a different country and pick a random one.

I'm expecting Rails core translations in Japanese, Korean and Chinese. Feel free to email me your own file and let me know on what project you used Globalite.

will_paginate and merb

Matt Aimonetti

Mislav & Pjhyett's will_paginate plugin has been a reference in the Rails world for a little while now. The need for Pagination is arguable, but that's not the point of this post.

There are few existing pagination solutions for Merb, but people moving from Rails and porting a Rails app will be happy to hear that will_paginate is now Merb compatible (DataMapper and ActiveRecord supported).

Mislav worked hard making sure that his next release will be framework agnostic and hopes to support both Rails 2.2 and Merb 1.0. You can see where is at by checking on his agnostic branch .

I did some work over the week end to make sure I could use will_paginate in one of my project. You can see my work there . I also built a temp gem for you to test WP with Merb until mislav releases the final version of the gem (especially if you are lazy and don't want to deal with git).

attachment_fu updated

Matt Aimonetti

I recently bugged Rick Olson so much about attachment_fu that he gave me SVN access to fix few bugs.

Rick being really busy with ActiveReload he didn't spend too much time maintaining attachment_fu .

On my side of things, I've been using attachment_fu on a lot of projects and I've been fixing bugs and adding new features.

My first contribution to attachment_fu is a fix for the ImageScience processor.

Attachment_fu is very flexible and let you use your favorite image processor:

RMagick based on ImageMagick and GraphicsMagick (known to leak memory and being a pain to setup)

minimagick based on ImageMagick

ImageScience based on FreeImage .

Like many rubyists, I like ImageScience for its simplicity and efficiency. However, attachment_fu had few problems when being used with ImageScience .

File sizes for thumbnails were not saved correctly in the database. Fixed

Thumbnails based on a gif files were not processed properly. So, this was the big problem. FreeImage has issues dealing with resizing gif files because of the gif palette limitation (256 colors). to avoid this problem thumbnails of

gif files are converted to png. However the thumbnail content type info in the database was not saved properly. That's now fixed.

Because of the gif bug reported above, any thumbnail link was broken since it was trying to link to the thumbnail version with a gif extension instead of a png one. Fixed

I also fixed a small bug related to S3 storage and the fact that a_fu had issues loading the config file. (Fixed)

I'll also try be able to add some of the S3 features I've been working on. As well as maybe enhancing the validation process.

In the mean time, you might want to read this blog post about better validation with attachment_fu.

If you are heavily using attachment_fu or starting using it and think that a google group would be great idea, please let me know in the comment and I'll try to convince Technoweenie that we need to set that up :)

Ooohh I almost forgot, if you fixed some bugs you found while using a_fu, please contact me so we can get a_fu bug free.

news update

Matt Aimonetti

I realized I haven't updated this blog in a while. Here is a quick update on what's happened and on things to come:

RailsConf 08 . Great conference, probably my last Rails Conf though. I'll be in Orlando for Ruby Conf 08 and I'll focus on 1 or 2 local conferences (probably mountain west and another one).

MerbCamp 08 in San Diego this Fall organized by SD Ruby . Details are not finalized yet but Yehuda Katz announced it during his Merb talk at RailsConf.

Moved this blog to a new Joyent accelerator with git support and finally have the possibility to use Ambition! (planning on moving from Mephisto to Feather)

Launched a client's Merb app and getting around 3 million hits/day. Merb is just awesome. (more info when the client's app gets out of beta)

I'll join Gregg Pollack from <https://railsenvy.com/> during Qcon and take part in the Ruby for the Enterprise track. My talk will focus on Merb usage in real life.

Renamed my github username, new repo url: <https://github.com/mattetti> (sorry about that)

Claude Can Now Take Control of Your Mac

John Gruber · Daring Fireball

Claude:

In Claude Cowork and Claude Code, you can now enable Claude to use your computer to complete tasks. When Claude doesn't have access to the tools it needs, it will point, click, and navigate what's on your screen to perform the task itself. It can open files, use the browser, and run dev tools automatically — with no setup required.

This feature is now available in research preview for Claude Pro and Max subscribers. It works especially well with Dispatch, which lets you assign Claude tasks from your phone.

I think you're nuts if you try this on your actual Mac, with all your actual data and files. But I thought people were nuts for using a lot of bleeding edge AI features before I tried them myself. It's certainly notable that Anthropic has shipped agentic AI on the Mac before Apple has, after Apple originally promised it to arrive a year ago.

The Claude Mac client itself remains a lazy Electron clunker. If Claude Code is so good I don't get why they don't prove it by using it to make an even halfway decent native Mac app.

See also: Techmeme .

★

sexy chart the video

Matt Aimonetti

A bit more than a month ago I posted a tutorial on how to use Flash with Rails to create some awesome/sexy graphs .

Since a lot of people seemed interested by the topic, the SDRuby guys asked me to do a intro talk on how to create Sexy Charts with super sexy Rails.

In the mean time, a lot of people were asking for a example app to look at. People knowing me know that I'm quite lazy and I don't like repeating tasks. I therefore decided to kill 2 birds with one stone and wrote a demo app that I would use during my presentation

As I was writing the demo app, I quickly realized that my talk would be even sexier if I would show some best practices. After all, an introduction talk is meant to help newbies learning the tricks that will change them in ninjas!

Sexy charts are sexy now, but in 15 years they might not look so sexy anymore. However BDD is super hot now and will always be sexy! (even though we'll probably adopt other even hotter approaches).

Based on the circumstances I decided that Sexy charts would become an excuse to show people how to do BDD using RSpec and how to test a XML view as described in this previous post

During my presentation I totally forgot to show people what what kind of XML

we were trying to feed am-Charts, so here is the file:

<https://pastie.caboo.se/120055>

The code used in the presentation is also available here

Presentation available here (the sound is a bit saturated, sorry about that. Note that we made the video big enough so you can follow with the code if you don't understand my accent :))

Feel free to watch the other SDRuby podcasts or even better, subscribe to our feed .

Next SDRuby meeting will be Thursday, December 6 @ 7:30pm

Location: UCSD CS Building

We'll be talking about Unobtrusive Javascript , the Facebook API , and hosting our first Rails Roundtable.

Newbies and experts welcome!

After all, an introduction talk is meant to help newbies learning the tricks that will change them in ninjas! Sexy charts are sexy now, but in 15 years they might not look so sexy anymore.

Matt Aimonetti

MacOS 26.4 Adds ‘Slow Charger’ Indicator for MacBooks

John Gruber · Daring Fireball

Tim Hardwick at MacRumors:

macOS Tahoe 26.4 includes a new slow charger indicator that tells MacBook users when their charging setup isn't delivering full power. As described in an updated Apple support document, a "Slow Charger" label now appears in orange text in the battery status menu and above the Battery Level graph in Battery settings. The indicator is accompanied by an info button for more details.

Apple says that to charge more quickly, users should use a power adapter and cable that deliver at least the minimum wattage recommended for their MacBook model.

This might be especially useful in Europe, where MacBooks no longer come with power adapters. Regular people often have no idea how power adapters work, and presume one charger is as good as another, if it works at all. After I posted this item back in October about the new MacBook Pros not shipping with chargers anywhere in Europe (not just the EU, even though it's an EU law that requires products to be available without included chargers), a bunch of readers regaled me with tales of a family member complaining about their MacBook losing battery life even while plugged in, only to discover that they were using wimpy 5- or 10-watt USB-C adapters.

★

latest merb and rails 30 news

Matt Aimonetti

Foy Savas, author of the "merb way" wrote a very well written post on "Rails monoculture"

Pat Eyler, wrote an article asking if Rails and Merb would be better together.

Ben Aldred, tell people to stop worrying and start loving Rails 3.

fotonauts.com a Rails and Merb Photo website developed by an ex-apple team, was featured in TechCrunch. Fotonauts is the perfect example of an app that will gain a lot from the merge.

Yehuda has been blogging a lot about every single step and even though most people are enjoying a well deserved break, you can read the details of the work started on the merge.

Basically, the work started, we are getting familiar with the rails code base and are optimizing things slowly but surely with a focus on testing JRuby. The Merb router is being optimized and ported over to Rails 3.0. Rails and Merb developers will be able to stick to their DSL (so we stay backward compatible). Merb bootloader is also being ported over without breaking the backward compatibility. Finally, ActiveSupport is being broken down in more manageable/independent chunks so people will be able to pick only what they want to use. A "mini" version is also on the work.

Merb 1.0.7 got released yesterday with a bunch of bug fixes.

merb_sequel 1.0 should be released sometime this week and i'm planning on adding rails i18n syntax support to merb_babel.

books to read in 2014

Matt Aimonetti

I like to read and I have a great community on Twitter so I asked them what they think others and myself should read in 2014. Here is the compiled list of the recommended books. The good news is that most of these books are under \$10, so why not picking a couple for the new year?

A big thanks to all the people who recommended these books, I'm looking forward to 2014!

Recommended by

@aweissman

@Adkron

The Orphan Master's Son (Pulitzer Prize for Fiction)

@bryanwoods

@aweissman

Mr. Penumbra's 24-Hour Bookstore

@conrey

@aweissman

@olg

@aweissman

Killing Floor (Jack Reacher, No. 1)

@drnic

Jonathan Strange and Mr Norrell

@olg

Thinking, Fast and Slow

Psychology / Business Decision-Making

@atduskgreg

The Signal and the Noise: Why So Many Predictions Fail-but Some Don't

Business Planning & Forecasting

@roidrage

The Life You Can Save: Acting Now to End World Poverty

Philanthropy & Charity

Theresa Preston-Werner

@tpdubs2

Thinking in Systems: A Primer

@roidrage

@kstewart

NOS4A2

@diminish7

Basic Economics: A Common Sense Guide to the Economy, 4th Edition

@rubiety

If you want more, you can check the previous book recommendation list

The Information: 'Apple Can "Distill" Google's Big Gemini Model'

John Gruber · Daring Fireball

Jessica E. Lessin, Amir Efrati, and Erin Woo, reporting for the paywalled-without-gift-links The Information:

While we have reported that Apple can tweak, or fine-tune, a version of Google's Gemini AI so that it responds to queries the way Apple wants, the agreement gives Apple a lot more freedom with Google's tech.

In fact, Apple has complete access to the Gemini model in its own data center facilities. Apple can use that access to produce smaller models that power specific tasks or are small enough to run directly on Apple devices so they can run the tasks faster, said a person who has direct knowledge of the arrangement.

The process of producing such models is called distillation , which essentially transfers knowledge from one large language model, which acts like a teacher, to another model that acts as a student.

That Apple negotiated this level of access is interesting, but not surprising. The biggest tell that this deal runs much deeper than simple white-labelling is that Apple will — or at least has the right to — run these Gemini-based models in Apple's own Private Cloud Compute datacenters.

★

problems with urls in merb head

Matt Aimonetti

I actually run into a small problem when updated an older Merb app. Here was how my router looked like:

```
Merb:: Router.prepare do
|r| r.resources :channels
do |channels|
channels.resources :shows
do |shows|
shows.resources :episodes
end end end
```

But after updating to the latest version of Merb, I got links looking like:

```
https://localhost:4000/
channels/# /shows
```

The first thing to do is to read Carl's wiki about the latest Router changes .

Carl explains that things got cleaned up in the router code and my routes should now look like:

```
Merb:: Router.prepare do
|r| r.resources :channels
do resources :shows do |
shows| resources :episodes
end end end
```

However that won't be enough.. You see my url used to look like that:

Now I can simplify it to:

```
url(:channel_shows, channel)
```

That still won't fix the problem, since the real problem comes from the fact that I was on Merb HEAD but not DataMapper HEAD. Updating DM clears things up. That's the price to pay to be on HEAD ;)

FYI the problem comes from the fact that DM doesn't add a `to_params` method to its objects. Rails users might recognize that

method used to convert an object into a string to create a route, something not really ORM agnostic and frowned upon by the DM/Merb teams.

Merb lets you specify the param to use for your routes using the `identify` method. Read Carl's wiki page for more cool stuff and see how to create some cool stuff like url slugs etc..

Note that even if you are using `ActiveRecord`, you'll need to update `merb_activerecord` as the new `identify` rules were updated in the ORM plugins.

More details about the vivo T5 Pro leak including its price and launch time frame

GSMarena

vivo is working on the T5 Pro, the successor to the T4 Pro, and it's expected to launch in India soon. The T5 Pro surfaced in an IMEI database back in February, and a few days ago its key specs got outed by a tipster.

Today, a new report claims the phone will become official in India by mid-April. It allegedly comes with a 6.8-inch AMOLED screen with "1.5K" resolution and 144Hz refresh rate, and it's powered by the Snapdragon 7s Gen 4 SoC. There's a 9,020 mAh battery keeping the lights on, which charges at 90W.

vivo T4 Pro

On the rear there's a 50MP main camera using Sony's IMX882...

*resources :channels do resources :shows do |shows| resources :episodes end end
end However that won't be enough.*

Matt Aimonetti

larubyconf 2012

Matt Aimonetti

During LA RubyConf 2012 in Los Angeles, CA Matt Aimonetti gave a talk entitled Ruby: time to move on .

##Description of the talk:

Let's be honest, Ruby became mainstream a few years back and it isn't the cool underground programming language it once was. It's quite likely that your cousin's boyfriend who's "into computers" knows what Ruby on Rails is. There are hundreds of books, conferences, training and meetups for Rubyists. Recruiters fight to hire whoever knows how to generate a scaffolded Rails app. But now cool kids can't stop talking about node.js, CoffeeScript, Clojure, Haskell and pushing code to the UI layer. What does it mean for the new, existing and prospecting Ruby developers? Is it time to jump ship and move on to something else?

##Slides

The slides are available on Matt's SpeakerDeck and can be [downloaded here]({{ page.slides }}).

##Video

```
{% video https://cdn.confreaks.com/system/assets/datas/3173/original/816-larubyconf2012-time-to-move-away-from-ruby-small.mp4 640 360 /images/matt_aimonetti_larubyconf2012_video.png %}
```

##Presentation website

"Michael Kennedy's Thoughts on Technology": Fire and Forget at Textual

Planet Python

If you read my Fire and Forget (or Never) about Python and asynchronous programming, you could think it's a super odd edge case. But a reader/listener, Richard, pointed me at Will McGugan's article The Heisenbug lurking in your async code . This is basically the same article, but in Will-style.

Will does say "This behavior is well documented, as you can see from this excerpt." True, but the documentation got this emphasis and warning in Python 3.12 whereas the feature `create_task` was added in Python 3.6/3.5 timeframe. So it's not just a matter of did we read the docs carefully. It's a matter of did we reread the docs carefully, years later?

Luckily Will added some nice concrete numbers I didn't have:

https://github.com/search?q=%22asyncio.create_task%28%22&type=code

This appears in over 0.5M separate code files on GitHub . To be clear, not every search result for `create_task` uses the fire-and-forget pattern, but just on the first page of results there are 5 instances.

If the design pattern to fix this is to:

Create a global set

When a task is added to the event loop, add it to the set

Remove it from the set when it's done

Wouldn't it have been better for the Python team to add this to the event loop internally once and solve this problem for everyone globally across the entire Python ecosystem?

It doesn't look like that's going to happen. So make sure you double check your code for `create_task` . And don't let the Heisenbugs bite.

And yes, I know about task groups . Several people told me that we could use task groups to hang on to the task. Yes, that's true. But task groups are incongruent with the fire-and-forget design pattern. Why? Because you create the group in a context manager and then you wait for all the tasks in the group to be finished. That doesn't allow you to fire off a task and then continue working. So task groups may or may not have fixed Will's problem, but they don't solve the one I was originally talking about .

ANALYSIS

IN BRIEF

Darknet, Darknet (security news): SmbCrawler is a credentialed SMB share crawler for red teams that discovers misconfigured shares and hunts secrets across Windows networks.

The Hacker News, The Hacker News (cybersecurity): A recently disclosed critical security flaw impacting Citrix NetScaler ADC and NetScaler Gateway is witnessing active reconnaissance activity, according to Defused Cyber and watchTowr. The vulnerability, CVE-2026-3055 (CVSS score: 9.3), refers to a case of insufficient input validation leading to memory overread, which an attacker could exploit to leak potentially sensitive information. Per

The Hacker News, The Hacker News (cybersecurity): TeamPCP, the threat actor behind the recent compromises of Trivy and KICS, has now compromised a popular Python package named litellm, pushing two malicious versions containing a credential harvester, a Kubernetes lateral movement toolkit, and a persistent backdoor. Multiple security vendors, including Endor Labs and JFrog, revealed that litellm versions 1.82.7 and 1.82.8 were published on March

Inaara Thawer, Inhabitat: Kaiserstraße is a new residential building currently under construction in Blumenau, Brazil. The project is designed by Alencar Arquitetura and aims to harness connections between residents and the natural world.

Planet Python, Planet Python: With the mountains of Python code that it's possible to generate now, how's your code review going? What are the limitations of human review, and where does machine review excel? Christopher Trudeau is back on the show this week with another batch of PyCoder's Weekly articles and projects.

Matt Aimonetti, Matt Aimonetti: Globalite contributor: Marcus Derencius just submitted a Portuguese translation for Rails.

Your application can now speak Portuguese thanks to Globalite and Marcus :)

Jane Street Blog, Jane Street Blog: At the end of last year, we decided to try something new: a challenge that would run alongside Advent of Code , where we asked the community to show us how they could design hardware to solve the same problems. We had no idea what level of participation to expect, but we received a huge number of submissions, many of which were incredibly creative!

Jane Street Blog, Jane Street Blog: At Jane Street, for the last several years, we have been increasingly interested in machine learning and its many use cases. This is why it was exciting when earlier this year myself and a few of my colleagues had the opportunity to attend the AAAI 2019 conference. We'd like to take this space to share with you some of the interesting projects and themes we saw at the conference.

Jane Street Blog, Jane Street Blog: We recently restructured our standard libraries at Jane Street in a way that eliminates the difference between Core_kernel and Core and we're happy with the result. The new layout should reach the open source world before the end of the year.

Jane Street Blog, Jane Street Blog: At Jane Street, we have always been heavy users of pre-processors, first with camlp4 and now ppx. Pre-processing makes the infrastructure a bit more complex, but it save us a lot of time by taking care of a lot of tedious boilerplate code and in some case makes the code a bit prettier.

Laura Cowan, Inhabitat: With climate change ramping up and an El Nino weather pattern in effect for the western hemisphere, heat island effect has become a real problem for cities sweltering under the summer sun. GAF created a product called Streetbond Pavement Coating that aims to turn that heat island effect around by reflecting heat back into space.

Shilpa Varma, HubSpot Product Blog:

On February 25, 2026, HubSpot experienced a service disruption affecting user access to specific types of workflows within our user interface. All backend workflow automations continued to execute normally without interruption - no automations were delayed, missed, or lost. We have completed a thorough analysis of this incident and are implementing targeted safeguards to prevent this class of failure from recurring.

Laura Cowan, Inhabitat: The Casitas at The Halles by Hive 3D and Eco Material Technologies are a new project to create sustainable, 3D-printed homes for vacation rentals near Austin, Texas. These unique low-profile homes use a technology that creates 92% less emissions than traditional concrete. Round Top, Texas, 80 miles east of Austin, will now be home to the world's first near zero-carbon, 3D-printed neighborhood.

Jane Street Blog, Jane Street Blog: We recently ran across a strange higgs-bugson that manifested itself in a critical system that stores and distributes the firm's trading activity data, called Gord. (A higgs-bugson is a bug that is reported in practice but difficult to reproduce, named for the Higgs boson , a particle which was theorized in the 1960s but only found in 2013.) In this post I'll walk you through the process I took to debug it. I tried to write down relevant details as they came up, so see if you can guess what the bug is while reading along.

Darknet, Darknet (security news): Dark web search engines in 2025 and how enterprises use monitoring, APIs and IOC hunting to detect credential leaks, impersonation and supply chain exposure.

KC Morgan, Inhabitat: Among the oldest tools that human hands ever created are axes, used to chop wood for fires and buildings. Wood is truly the mother of sustainable resources, a material that will grow back again and again. And now, wood is becoming the "it" design material. Sweden is starting an ambitious new project that will demonstrate the majesty of wood, and the possibilities it represents in sustainable building.

Wide Monitor

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